Workplan for October

Activities: What is the oxidation number of oxygen in most of its compounds?

Activity code: 1.1

Start date: 12/10/2026 **End date: 17/10/2026**

A. +1
B. -1
C. -2
D. +2
Answer: C

2. What is the oxidation number of hydrogen in most compounds?

A. +1
B. -1
C. +2
D. -2
Answer: A

3. What is the oxidation number of sodium in NaCl?

A. -1
B. 0
C. +1
D. +2
Answer: C

4. An increase in oxidation number during a reaction indicates?

A. Reduction
B. Oxidation
C. Neutralisation
D. Hydrolysis
Answer: B

Activity code: 1.2

Activity description: Review of exam questions?

Start date: 20/10/2026 **End date: 25/10/2026**

A. Oxidation
B. Combustion
C. Reduction
D. Sublimation
Answer: C

6. Which of the following is a reducing agent?

A. Oxygen
B. Chlorine
C. Hydrogen
D. Potassium permanganate
Answer: C

7. Electrolysis is the process by which?

A. Electricity is produced from chemicals only
B. A chemical compound is decomposed using electricity
C. Water is purified by filtration
D. A gas is compressed
Answer: B

8. In an electrolytic cell, oxidation takes place at the?

A. Cathode
B. Anode
C. Electrolyte
D. Salt bridge
Answer: B

9. In an electrolytic cell, reduction takes place at the?

A. Anode
B. Cathode
C. Electrolyte
D. Switch
Answer: B

10. During electrolysis, positively charged ions move towards the?

A. Anode
B. Cathode
C. Electrolyte surface

D. Salt bridge
Answer: B

11. During electrolysis, negatively charged ions move towards the?

A. Cathode
B. Anode
C. Battery terminal only
D. Salt bridge
Answer: B

12. Which of the following is a good electrolyte?

A. Sugar solution
B. Sodium chloride solution
C. Kerosene
D. Pure oil
Answer: B

13. Which substance is produced at the cathode during the electrolysis of acidified water?

A. Oxygen
B. Chlorine
C. Hydrogen
D. Nitrogen
Answer: C

14. Which gas is produced at the anode during the electrolysis of acidified water?

A. Hydrogen
B. Oxygen
C. Nitrogen
D. Carbon dioxide
Answer: B

15. What is the main purpose of electroplating?

A. To increase the boiling point of a metal
B. To coat an object with a thin layer of another metal
C. To melt metals
D. To remove all impurities from water
Answer: B

16. Which metal is commonly used to electroplate iron objects to prevent corrosion?

A. Zinc
B. Gold
C. Silver
D. Copper
Answer: A

17. Corrosion of iron is commonly known as?

A. Tarnishing
B. Rusting
C. Cracking
D. Melting
Answer: B

18. Rusting of iron requires?

A. Nitrogen and hydrogen
B. Oxygen and water
C. Carbon dioxide only
D. Hydrogen and chlorine
Answer: B

Activity: Marking exam questions

Code: 1.3

Start date: 26/10/2026 **End date: 30/10/2026**

A. Keeping iron in moist air
B. Painting the iron surface
C. Adding water
D. Scratching the surface
Answer: B

20. An alloy is?

A. A pure metal
B. A mixture containing a metal and one or more other elements
C. A non-metallic compound
D. A gas dissolved in water
Answer: B

21. Brass is an alloy of?

A. Copper and zinc
B. Copper and tin
C. Iron and carbon
D. Aluminium and magnesium
Answer: A

22. Bronze is mainly an alloy of?

A. Copper and zinc
B. Copper and tin
C. Iron and carbon
D. Sodium and potassium
Answer: B

23. Steel is mainly an alloy of?

A. Copper and zinc
B. Iron and carbon
C. Aluminium and copper
D. Zinc and tin
Answer: B

24. Which property makes aluminium useful for making aircraft?

A. It is very dense
B. It is light and relatively strong
C. It is a poor conductor
D. It rusts rapidly
Answer: B

25. Which of the following is an important ore of aluminium?

A. Bauxite
B. Haematite
C. Galena
D. Limestone
Answer: A

26. Haematite is an important ore of?

A. Aluminium
B. Iron

C. Copper
D. Zinc
Answer: B

27. Galena is an important ore of?

A. Lead
B. Iron
C. Aluminium
D. Sodium
Answer: A

28. The extraction of a metal from its ore generally involves?

A. Concentration and reduction
B. Freezing only
C. Evaporation only
D. Filtration only
Answer: A

29. Which reducing agent is commonly used in the extraction of iron from iron ore in a blast furnace?

A. Carbon monoxide
B. Oxygen
C. Nitrogen
D. Hydrogen chloride
Answer: A

30. Limestone is added to a blast furnace mainly to?

A. Produce oxygen
B. Act as a flux and remove impurities
C. Produce hydrogen
D. Increase the density of iron
Answer: B

31. Which gas is produced when limestone decomposes in a blast furnace?

A. Hydrogen
B. Oxygen
C. Carbon dioxide

D. Nitrogen
Answer: C

32. What is the chemical formula of limestone?

A. CaO
B. $CaCO_3$
C. $CaCl_2$
D. $Ca(OH)_2$
Answer: B

33. What is the chemical formula of quicklime?

A. $CaCO_3$
B. CaO
C. $Ca(OH)_2$
D. $CaCl_2$
Answer: B

34. What is the chemical formula of slaked lime?

A. CaO
B. $CaCO_3$
C. $Ca(OH)_2$
D. $CaCl_2$
Answer: C

35. Which organic compound contains the functional group –OH?

A. Alkane
B. Alcohol
C. Alkene
D. Alkane acid
Answer: B

36. What is the molecular formula of ethanol?

A. CH_3OH
B. C_2H_5OH
C. C_2H_4
D. C_2H_6
Answer: B

37. Which functional group is present in carboxylic acids?

A. –OH
B. –COOH
C. –CHO
D. $–NH_2$
Answer: B

38. Ethanoic acid is commonly found in?

A. Vinegar
B. Petrol
C. Limestone
D. Soap powder
Answer: A

39. What is formed when an alcohol reacts with a carboxylic acid?

A. Alkane
B. Ester
C. Alkene
D. Amine
Answer: B

40. The reaction between an alcohol and a carboxylic acid to form an ester is called?

A. Hydrolysis
B. Esterification
C. Polymerisation
D. Neutralisation
Answer: B

41. Which of the following is an alkene?

A. Methane
B. Ethane
C. Ethene
D. Ethanol
Answer: C

42. What is the general formula of alkenes?

A. C_nH_{2n+2}
B. C_nH_{2n}
C. C_nH_{2n-2}

D. C_nH_{n+2}
Answer: B

43. Which reagent can be used to test for an unsaturated hydrocarbon?

A. Limewater
B. Bromine water
C. Sodium hydroxide
D. Benedict's solution
Answer: B

44. What happens to bromine water when it reacts with an alkene?

A. It turns blue
B. It becomes colourless
C. It turns green
D. It produces a white precipitate
Answer: B

45. Polymerisation is a process in which?

A. Large molecules break into atoms
B. Small molecules join together to form large molecules
C. Metals react with acids
D. Salts dissolve in water
Answer: B

46. What polymer is formed from ethene?

A. Polyethene
B. PVC
C. Nylon
D. Terylene
Answer: A

47. Which of the following is a natural polymer?

A. Polyethene
B. PVC
C. Cellulose
D. Polystyrene
Answer: C

48. Which of the following is a greenhouse gas?

A. Carbon dioxide
B. Oxygen
C. Nitrogen
D. Hydrogen
Answer: A

49. Which process is mainly responsible for the production of oxygen by green plants?

A. Respiration
B. Photosynthesis
C. Fermentation
D. Combustion
Answer: B

50. Which of the following is a major component of natural gas?

A. Methane
B. Ethene
C. Ethanol
D. Ethanoic acid
Answer: A